



Your project title

BSc. in Software Engineering

Your name

Supervisor

Date:

Table of Contents

Must start on a new page

List of Tables

Must start on a new page

List of Figures fig1

Must start on a new page

List of Acronyms/Abbreviations

Must start on a new page

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ABSTRACT

One continuous paragraph that captures the introduction, Problem statement, methodology, and anticipated results in numerical format, and hypothesis
Must be 200-250 words max

- Use Times New Roman Font size 12
- Use Justified alignment
- Spacing is 1.5
- Citation style APA

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CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Throw in some statistics here e.g “In Africa Malaria Accounts for 40% of the population. This number is further magnified in country A (citation inserted here)”, Geographical setting,

New paragraph: traditional approaches in addressing what you are talking about in pgh 1 and software approaches

1.2 Problem statement

- Must be well-cited
- Must have 3 paragraphs, each representing a sub-problem, well cited
- Must have the last paragraph stating that this work introduces a solution that contains That you aim to implement in order to address the above problems

1.3 Project’s main objective

What is the overall aim of your project?

E.g. To develop a software solution that addresses gaps mentioned in the problem statement in order to.....

1.3.1 List of the specific objectives

Break down the overall aim into 3 specific objectives:

1. To understand, review literature, collect data and insight
2. Develop a solution based on
3. Collect and verify results based on measurable metrics that address the gap. If you are getting rid of malaria, you want to show if the number of patients has reduced on the advent of your software, or the number of plasmodium-carrying mosquitoes reduced. You define the metrics based on the problem (this is crucial aside from technical metrics). At the end of the day, did you solve the problem?

1.4 Research questions

(.....)

1.5 Project scope

Geographically, targeting to test the product in an entire country within 3 months is not realistic,so what about trying it out with a sample of people in Bumbogo,or Kimironko and

using that as a sample of the population I want to develop the solution for? Realistic number of users, Realistic scale of the product technical-wise

1.6 Significance and Justification

What do you think will happen upon the successful implementation of your software? have short 2 1 - 3 line paragraphs. A maximum of 3 paragraphs

1.7 Research Budget

Your research might require services or tools/components that will be bought, feel free to add tabulate your budget in this section. Though it is good to have everything in budget but be encouraged to only budget for the things that are readily available (this that can only be paid for without an alternative)

1.8 Research Timeline

Use the Gantt chart to present how different components of the project will be implemented with respect to time.

CHAPTER TWO: LITERATURE REVIEW START WITH THIS CHAPTER

2.1 Introduction

In one paragraph answer the following what kind of literature were you on the hunt for (it must be software-related literature not just general literature)? What methodology did you use to review the literature? Did you sample papers? Which indexed platforms did you explore to find literature? How many papers/literature did you find?

2.3 Overview of Existing System

delete this while editing. You are reviewing the competition. Strictly review software-related systems. You want to check what software solutions exist around the area you are trying to tackle. You want to prove to the reader that there are software systems that appear to solve the problem but are lacking in some aspects that you want to address.

2.4 Review of Related Work

2.3.1 Summary of Reviewed Literature

2.5 Strengths and Weakness of the Existing System(s)

2.6 General comment and Conclusion

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

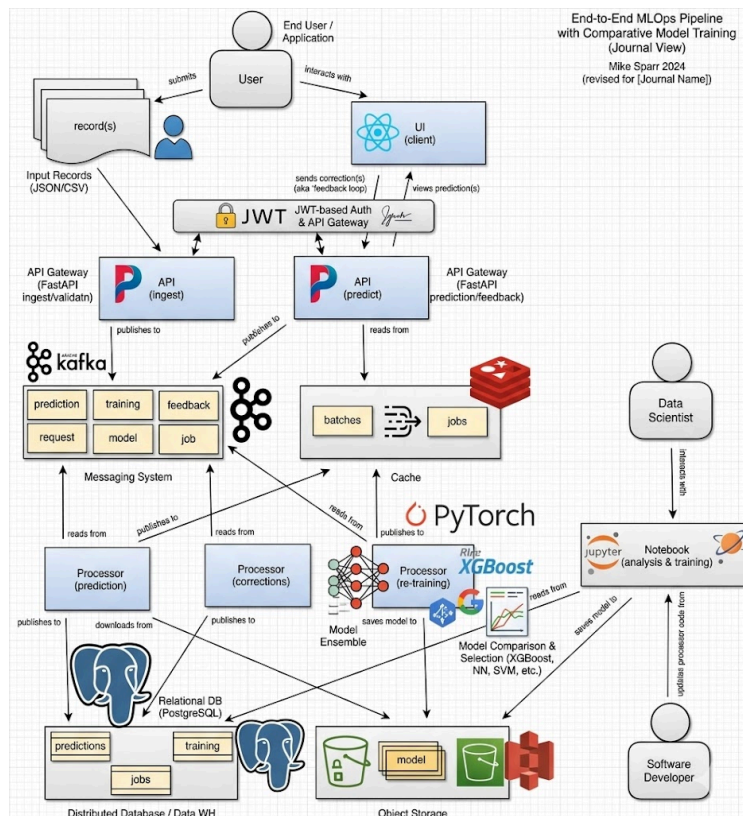
3.1 Introduction

An introduction of the approaches you will use to design. Be brief, concise and clear. The research methodologies description can go here.

3.2 Research Design (including the development model used)

Proposed model Diagram customized to your mission

3.4 System Architecture NB REFINE THE DIAGRAM PLEASE



4. System & Technology Architecture

The system is built on a modular, layered stack to ensure data flows efficiently from input to the backend.

- **Frontend/Input:** A Web Dashboard (e.g., React or Streamlit) for user interaction and data ingestion.
- **Backend:** A robust API layer (e.g., FastAPI or Flask) to handle logic and model requests.
- **Database Layer:**
 - * **PostgreSQL:** For structured relational data and user management.
 - **Vector Database (e.g., Pinecone/Milvus):** If managing embeddings or high-dimensional similarity searches.
- **Infrastructure:** Containerized deployment using Docker for consistent environment management.

1. Data Definition & Acquisition

- **Data Sources:** Clearly state if you are using a multi-source or single-source approach.
 - **Example:** "This project integrates three distinct sources: internal transaction logs, third-party demographic APIs, and historical CSV archives."

- *Data Volume: Specify the scale of the dataset (e.g., total rows, GB of storage, or timeframe).*
- *Citation & Provenance: Document the origin of each dataset and any licensing or privacy constraints.*

2. Machine Learning Training Pipeline

The pipeline ensures a repeatable path from raw data to an optimized output.

Pipeline Flow:

- *Input: Structured or unstructured raw data.*
- *Quality Phase: Includes data validation, outlier removal, and cross-validation to ensure the training set meets performance standards.*
- *Expected Output: A deployable inference model or a set of predictive insights.*

3. Model Comparison & Architecture

- *To identify the optimal solution, the project will benchmark the following algorithms for the classification task: XGBoost Vs Decision Trees Vs Neural Networks Vs SVM (Support Vector Machines) etc.....*
- *Compare Different Models for LLMs dont pick defaults without justification (LLama, Mistral, NLLB, Qwen)*
- *Compare different Policies for RL - You can use ML tech II findings here*

3.5 UML Diagrams

In this section, you must include:

- A use case diagram a class diagram
- Class Diagram
- ERD diagram, entity relationship diagram
- A sequence diagram is also mandatory

You must use the correct notations for each diagram, avoid generated diagrams, or if you're going to use AI-generated diagrams, please make sure you check whether it gives you the correct notations and the correct shapes for the respective diagrams

- **3.6 Development Tools**

References (APA Format)